

Problem Statement: A line editor is a command line text editor that works with one line at a time. The document is stored in memory that the user can insert, delete and display lines using commands.

Data Structure choice.

Data Structure: Dynamic Array of strings
char **lines;

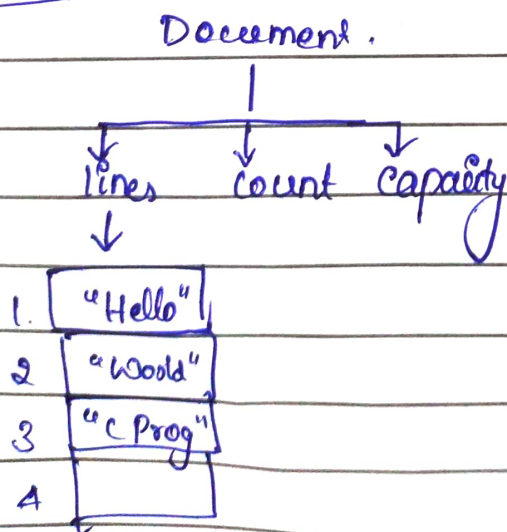
Document contains:

- lines → Stores pointers to each line
- count → current number of lines
- capacity → allocated number of lines slots

Why?

- Direct access to a line
- Easy to implement
- Suitable for a small document
- Can grow dynamically

Structure



Commands

insert <line number>

delete <line number>

display

save <filename>

load <filename>

search <word>

count

help

quit

INSERT

INSERT (linenumber, text)

1. Check if line number is valid.
2. If array is full, increase capacity.
3. Convert line number to array index.
4. Shift existing lines one position right.
5. Allocate memory for new line.
6. Copy text into new line.
7. Increase count.

DELETE

DELETE (linenumber)

1. Check if document is empty.
2. Check if line number is valid.
3. Convert line number to array index.
4. Free the selected line.
5. Shift remaining lines one position left.
6. Decrease count.

DISPLAY

DISPLAY()

1. Check if document is empty.

2. For every line:

print line number

print line text.